

From hyperspectral CHRIS to four RGBN bands

The cloud model eats **4 bands** (R, G, B, NIR). CHRIS has **18 to 62 bands** per mode. So each pseudo-band is built from the CHRIS bands that fall in it.

method 1 of 2, plain mean

which each CHRIS band goes to the Sentinel-2 band it overlaps the most

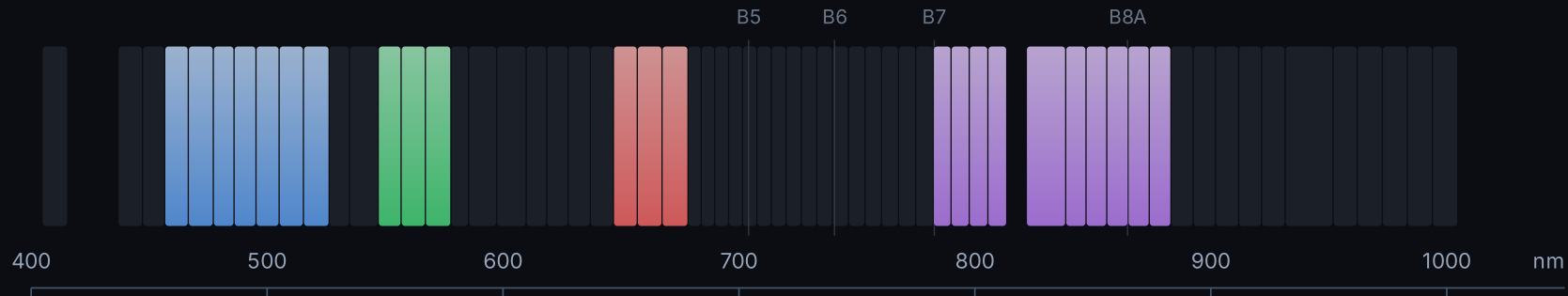
how plain **mean** of the CHRIS bands assigned to each pseudo-band

then DN/100 000 or TOA/10 000, clip to 5, and a per-tile z-score inside the network

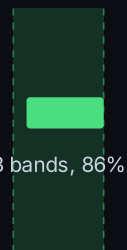
This is the training mapping, the one the model actually learned. Method 2, SRF-weighted resampling, starts after the mapping table.

Mode 1 : 62 bands, ~34 m

M1, mean



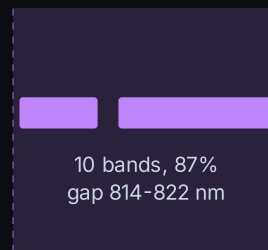
B2
459-525 nm



B3
542-578 nm



B4
649-680 nm



B8
780-886 nm

Mode 2 : 18 bands, ~17 m

M1, mean



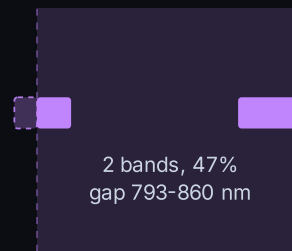
B2
459-525 nm



B3
542-578 nm



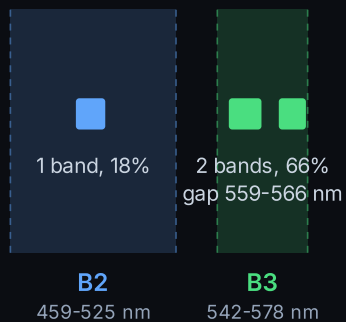
B4
649-680 nm



B8
780-886 nm

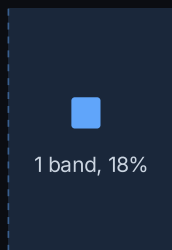
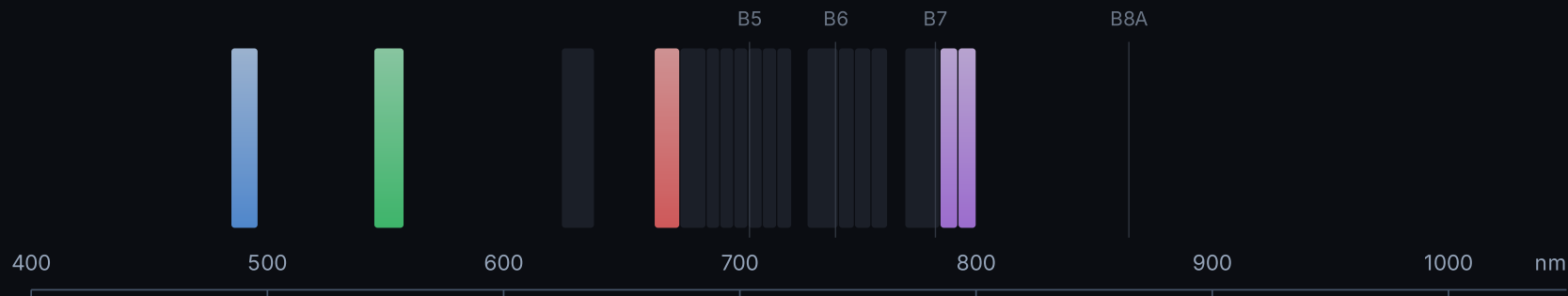
Mode 3 : 18 bands, ~13 m

M1, mean



Mode 4 : 18 bands, ~11 m

M1, mean



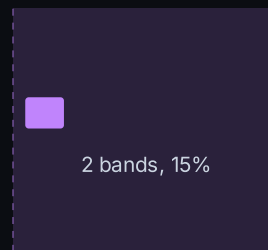
B2
459-525 nm



B3
542-578 nm



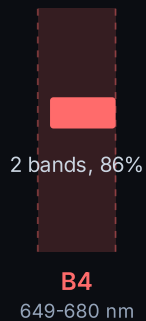
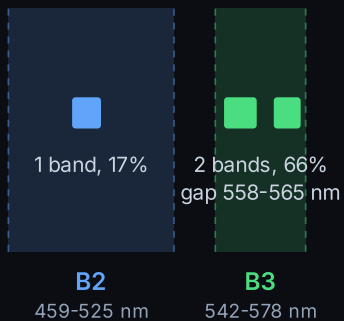
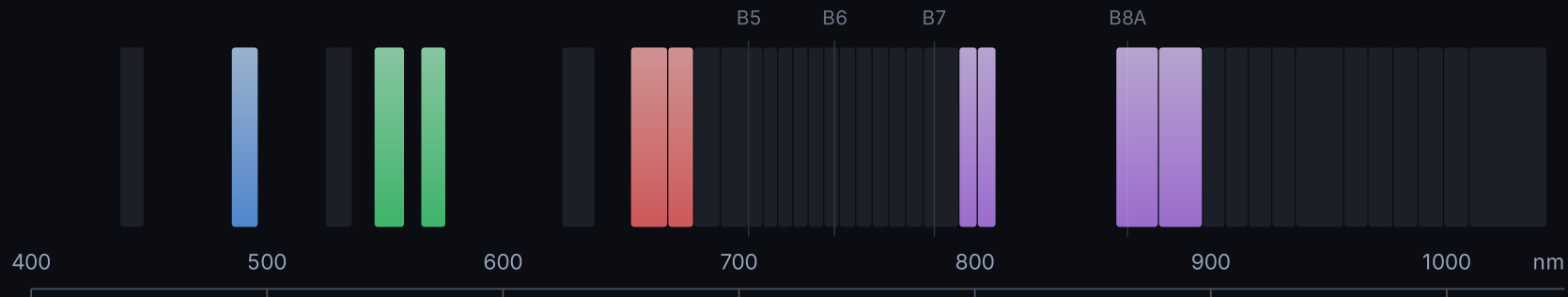
B4
649-680 nm



B8
780-886 nm

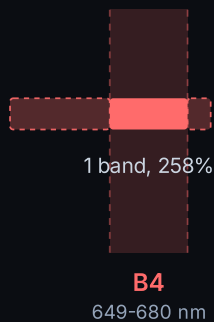
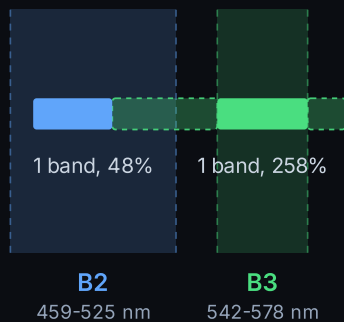
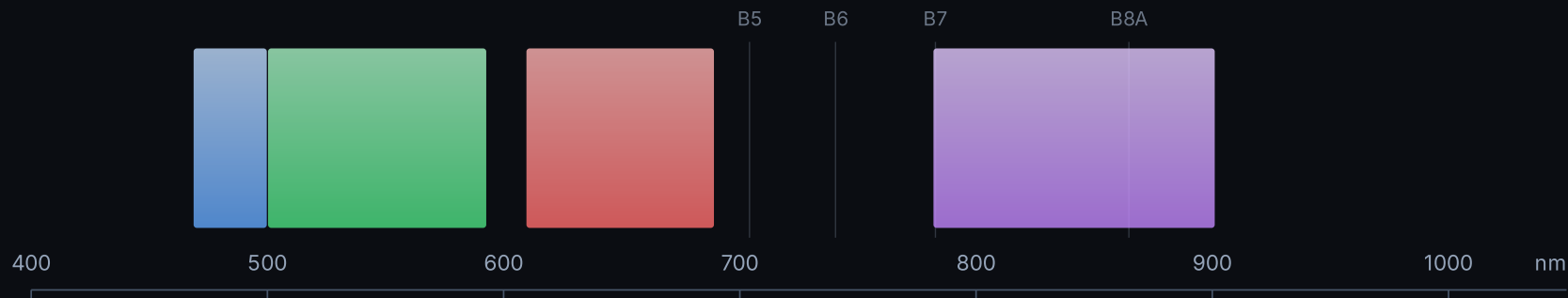
Mode 5 : 37 bands, ~20 m

M1, mean



Mode 20 : 4 bands, ~17 m, DN only

M1, mean



Method 1: where the bands end up

R, B4

G, B3

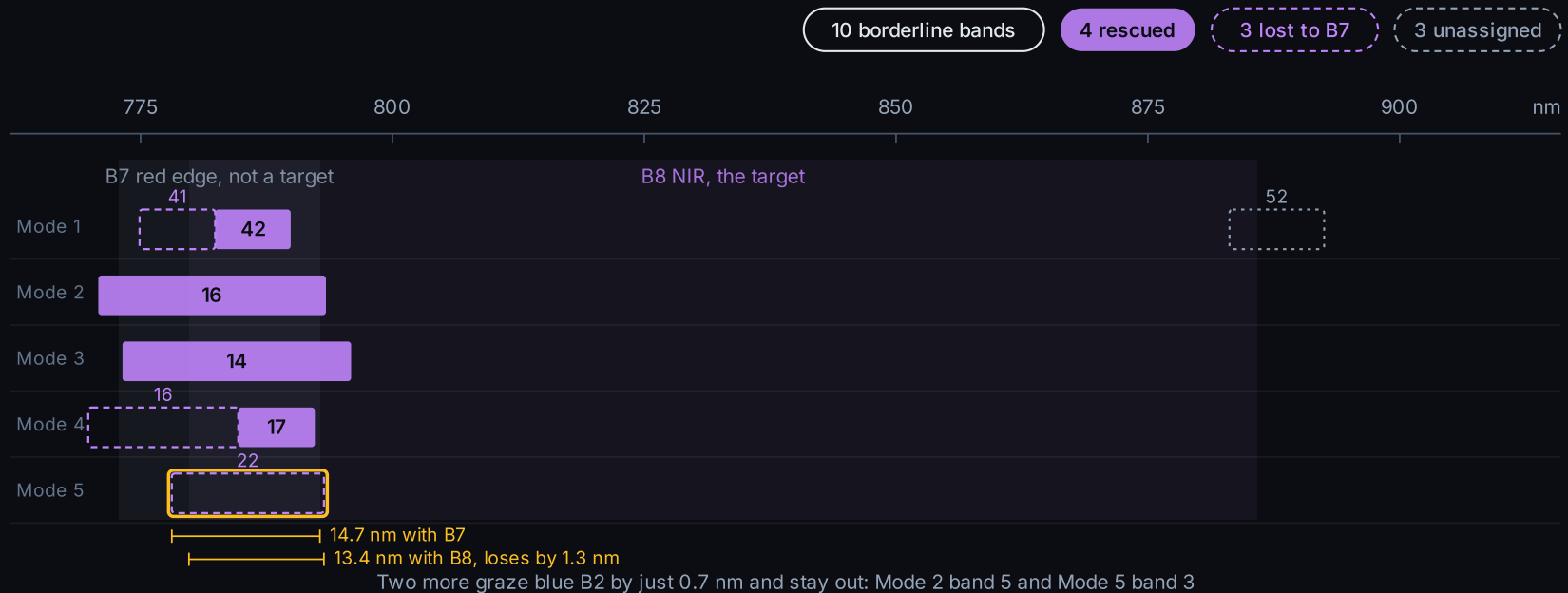
B, B2

NIR, B8

Mode	B2 blue	B3 green	B4 red	B8 NIR
1	4-10 (7), 457-526 nm	13-15 (3), 547-578	23-25 (3), 647-678	42-51 (10), 782-883
2	3-4 (2), 485-517	6-7 (2), 555-580	10-12 (3), 644-686	16-17 (2), 771-887
3	2 (1), 486-497	4-5 (2), 546-577	7-8 (2), 656-683	14-15 (2), 773-890
4	1 (1), 484-496	2 (1), 545-558	4 (1), 664-675	17-18 (2), 785-800
5	2 (1), 485-496	4-5 (2), 545-576	7-8 (2), 654-681	23-26 (4), 793-896
20	1 (1), 469-500	2 (1), 500-593	3 (1), 609-689	4 (1), 782-901

Training mapping. Numbers are 1-based CHRIS band indices; nm is the spectral span. Stack order: R, G, B, NIR.

What the max overlap rule leaves out



Each of these bands touches the NIR target but the winner takes all rule sends it away. Four were rescued by hand for training, three were left out after losing the red edge B7 duel (including Mode 5 band 22, the heaviest NIR band the SRF sees), and three stay unassigned under the 30 percent overlap threshold.

Method 2: SRF-weighted resampling

method 2 of 2, SRF weights

which every CHRIS band with nonzero S2A response takes part, no winner takes all

how w_i is the integral of the S2A response curve over the CHRIS band, and the pseudo-band is the weighted mean

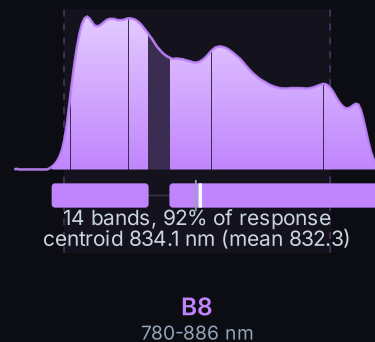
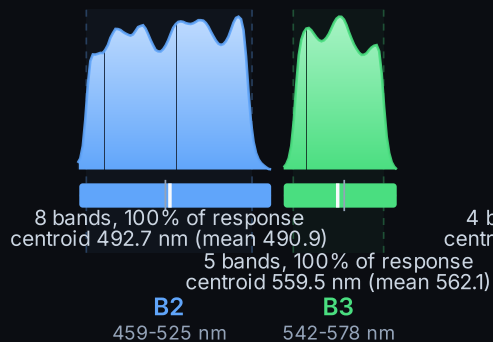
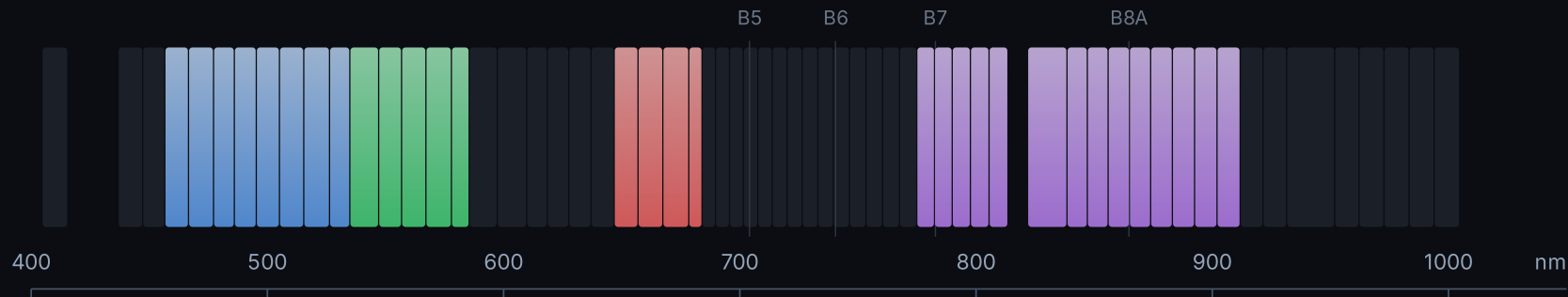
then weights are normalized by the area CHRIS actually covers, not by the full SRF area

$$\tilde{L}_g = \frac{\sum_i w_i L_i}{\sum_i w_i}, \quad w_i = \int_{\lambda_i^{lo}}^{\lambda_i^{hi}} R_g(\lambda) d\lambda$$

Same four targets, read through the real S2A response curves instead of flat FWHM boxes.

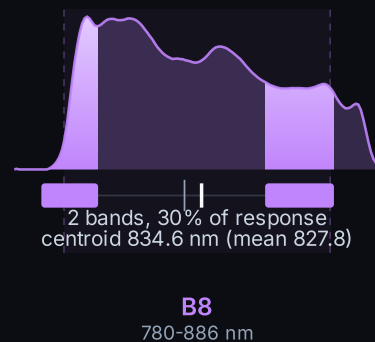
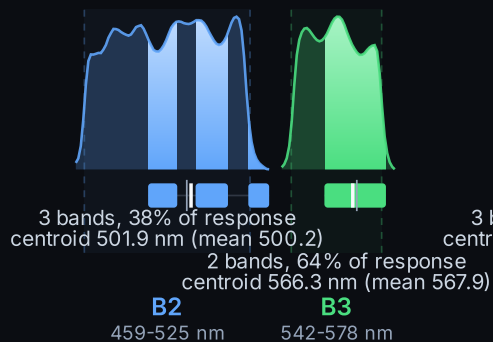
Mode 1: 62 bands, ~34 m

M2, SRF

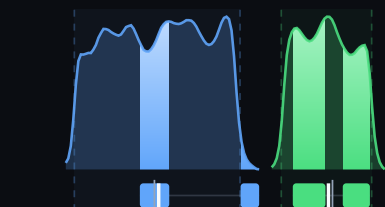


Mode 2 : 18 bands, ~17 m

M2, SRF



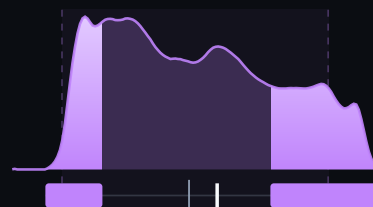
Mode 3 : 18 bands, ~13 m M2, SRF



2 bands, 18% of response
centroid 493 nm (mean 491.3)
2 bands, 65% of response
centroid 560.6 nm (mean 562.2)
B2
459-525 nm
B3
542-578 nm



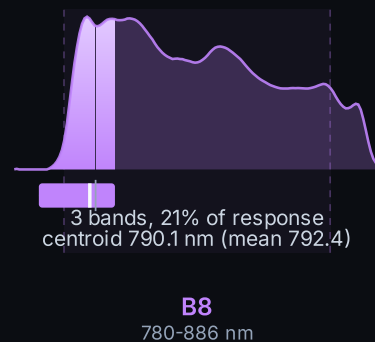
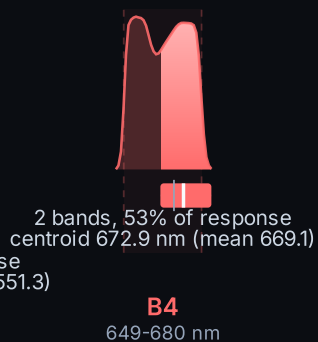
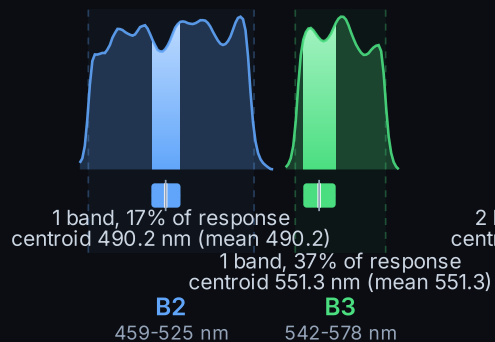
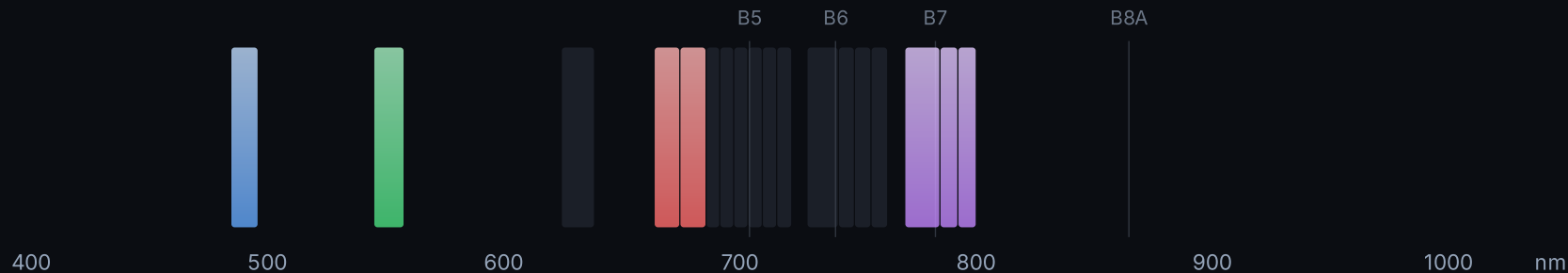
2 bands, 77% of response
centroid 668.6 nm (mean 670.4)
B4
649-680 nm



3 bands, 38% of response
centroid 841.6 nm (mean 830.4)
B8
780-886 nm

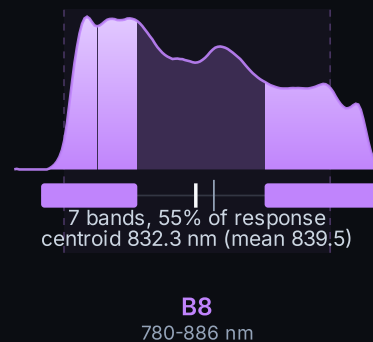
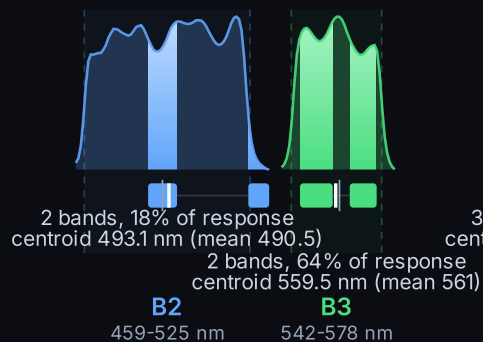
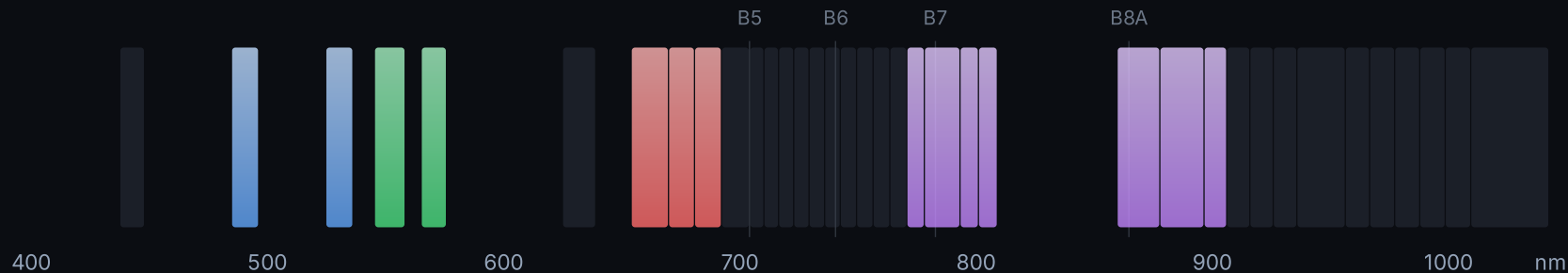
Mode 4 : 18 bands, ~11 m

M2, SRF



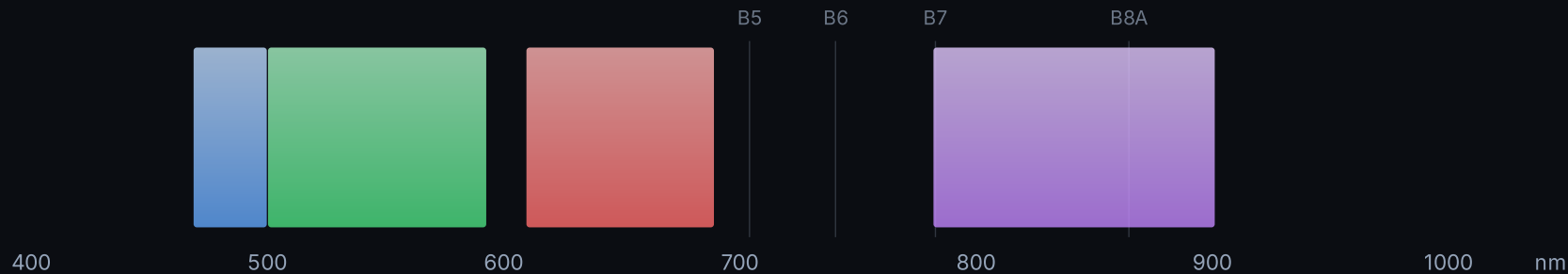
Mode 5 : 37 bands, ~20 m

M2, SRF



Mode 20 : 4 bands, ~17 m, DN only

M2, SRF



Band 2 feeds both blue and green: the blue centroid moves 25 nm toward the green.

Plain mean vs SRF weights

Mode	B2 blue	B3 green	B4 red	B8 NIR
1	7 → 8	3 → 5	3 → 4	10 → 14
2	2 → 3	2 → 2	3 → 3	2 → 2
3	1 → 2	2 → 2	2 → 2	2 → 3
4	1 → 1	1 → 1	1 → 2	2 → 3
5	1 → 2	2 → 2	2 → 3	4 → 7
20	1 → 2	1 → 1	1 → 1	1 → 1

CHRIS bands entering each pseudo-band. Left number is the plain mean set, right number the SRF set. Mode 5 band 22 comes back on its own, about a fifth of the B8 mix, and in mode 20 the SRF hands band 2 a share of the blue.